THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra

Dec 19, 2023 9:30 am – 12:00 noon

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the answer book.

Answer ALL SIX questions.

Notes:

- Write down your university number, seat number and model of calculator used on the cover of the answer book.
- The blank pages in the answer book can be used for rough work. You can also use them as additional answer space provided that you make a clear indication in the original answer space.
- Unless otherwise specified, you should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- Some formulae:

For $n \geq 2$:

$$\int \cos^n \theta \, d\theta = \frac{1}{n} \cos^{n-1} \theta \sin \theta + \frac{n-1}{n} \int \cos^{n-2} \theta \, d\theta$$

$$\int \sin^n \theta \, d\theta = -\frac{1}{n} \sin^{n-1} \theta \cos \theta + \frac{n-1}{n} \int \sin^{n-2} \theta \, d\theta$$

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1. (45 marks) For this question, give answers only. No explanation is needed.

Question (a)-(c). Let D be the bounded region in $\mathbb{R}^2$ lying inside the semi-ellipse $4x^2 + y^2 = 9$ where $x \geq 0$.

(a) Express the area of the semi-ellipse in terms of an integral in the order $dxdy$. (Do not evaluate the integral.)

(b) Express the area of the semi-ellipse in terms of an integral in the order $dydx$. (Do not evaluate the integral.)

(c) Find the volume under the graph of the function $f(x, y) = x$ over the region D.

Question (d)-(f). Let k be a real number, and let $A = \begin{bmatrix} 1 & 3 & 2 & 2 \\ -3 & 0 & 1 & 5 \\ 2 & k & 6 & 2 \\ 2 & 4 & 1 & 3 \end{bmatrix}$.

(d) Compute $\det A$ in terms of k.

(e) Compute $\det 2A$ in terms of k.

(f) Consider a homogeneous system of linear equations having A as the coefficient matrix. Find the value(s) of k for which the system has a unique solution.

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(g) Using ONE iteration of Newton-Raphson's method, solve the system of equations

$$\begin{cases} x^2 + 2y = 1 \\ x + y^2 = 3 \end{cases}$$

with the initial guess $(-2, -2)$. (Round your answer to 2 decimal places.)

(h) Suppose the matrix $\begin{bmatrix} 0 & 1 & a & 0 & 2 \\ 0 & 0 & 0 & b & 1 \\ 0 & 0 & 0 & 0 & c \end{bmatrix}$ is in reduced row echelon form. Find the values of a, b and c. Write down X if the value of a variable cannot be determined. For example, write down $c = X$ if c cannot be determined.

(i) Evaluate the double integral $\iint_R y^3 dxdy$, where R is triangular region with the vertices $(1, 2)$, $(0, 1)$, $(4, 0)$.

(j) Find the maximum value of the function $f(x, y, z) = x + 2z$ under the two constraints $x^2 + 2y^2 = 10$ and $x^2 + y^2 + 4z^2 = 8$.

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(k) Find the value of k such that $\begin{pmatrix} 3 \\ 2 \\ k \end{pmatrix}$ belongs to the span of $\left\{ \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}, \begin{pmatrix} -3 \\ -4 \\ 1 \end{pmatrix} \right\}$.

(l) It is known that $\begin{vmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 1 & 3 & 2 & 5 \\ 4 & 2 & 4 & 1 \end{vmatrix} = 4$. Compute $\begin{vmatrix} 1 & 3 & 2 & 5 \\ 405 & 206 & 407 & 108 \\ 1 & 2 & 3 & 4 \\ 8 & 4 & 8 & 2 \end{vmatrix}$.

(m) Let H be a solid sphere with radius a. Given that the density at each point of H is equal to the distance from this point to the boundary, write out the triple integral in spherical coordinates which can be used to find the mass of H. (Your answer should be a triple integral involving ρ, ϕ and θ. Do not evaluate the integral.)

(n) Write out the double integral which can be used to determine the area of the region that lies inside $r = 3 + 2\sin\theta$ and outside $r = 2$. (Do not evaluate the integral.)

(o) Consider the double integral $\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{0} \cos(x^2+y^2)\, dy\, dx$. If a change of variable is applied to transform the integral into one in polar coordinates, what will be the resulting integral? (Your answer should be a double integral involving r and θ. Do not evaluate the integral.)

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2. (18 marks) In each multiple choice question below, some numbered choices are given and some are correct (wordings like *is* and *are* do <u>not</u> indicate the singularity/plurality of the number of correct choices). Answer the question by adding up the numbers of the correct choices. In case all choices are wrong, answer 0. No explanation is needed.

EXAMPLE: Which of the following is correct?

(1) $1 + 2 = 3$.

(2) $1 + 1 = 2$.

(4) $1/0 = 1$.

(8) $1 - 1 = 0$. Answer: 11.

(a) Consider a solid cylinder in $\mathbb{R}^3$ described by the constraints $(x+1)^2+y^2 \leq 1$ and $0 \leq z \leq 4$. A point P lies strictly inside this cylinder. Suppose P has cylindrical coordinates (r, θ, z) and spherical coordinates (ρ, ϕ, θ) respectively. Which of the following descriptions is correct?

(1) $r = 1$

(2) $\rho = 1$

(4) θ can take any value from 0 to 2π (excluding 0 and 2π).

(8) ϕ can take any value from 0 to $\frac{\pi}{2}$ (excluding 0 and $\frac{\pi}{2}$).

(b) Which of the following are relative extreme points of $w = (3x+4x^3)(y^2+2y)$?

(1) $(0, 0)$

(2) $(0, -2)$

(4) $(2, 0)$

(8) $(-1, 1)$

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(c) Let $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ be linearly independent vectors in $\mathbb{R}^3$. Which of the following must be true?

(1) $\mathbf{x}$ and $\mathbf{y}$ are linearly independent.

(2) $\mathbf{x}$, $\mathbf{y}$, $\mathbf{z}$ and $\mathbf{w}$ are linearly independent for some vector $\mathbf{w}$ in $\mathbb{R}^3$.

(4) $2\mathbf{x}$, $3\mathbf{y}$ and $4\mathbf{z}$ are linearly independent.

(8) $\text{Span}\{\mathbf{x}, \mathbf{y}\} = \mathbb{R}^2$.

(d) Suppose the size of the augmented matrix of a system of linear equations is 4×4. Which of the following are possible?

(1) the system has no solution

(2) the system has a unique solution

(4) the system is consistent and there are exactly 2 free variables

(8) the system is consistent and the augmented matrix is invertible

(e) Let A be an $m \times n$ matrix. If the system $A\mathbf{x} = \mathbf{b}$ has a unique solution, which of the following is possible?

(1) $m < n$

(2) $m = n$ and A is invertible.

(4) $m = n$ and A is non-invertible.

(8) $m > n$

(f) Let A and B be $n \times n$ matrices with real entries. Which of the following must be true?

(1) $\det(A^2) \geq 0$

(2) $\det(A^3) = \det A$

(4) $\det(AB) = \det(BA)$

(8) $\det(A + B) = \det A + \det B$

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3. (7 marks) Find the absolute maximum and absolute minimum values of

$$f(x, y) = x^2 + y^2 + y + 1$$

subject to the constraints $x^2 + y^2 \leq 1$ and $x \leq 0$.

4. (10 marks) Evaluate $\iiint_D x^2 + y^2 dV$, where D is the portion of the ellipsoid $x^2 + \frac{y^2}{4} + \frac{z^2}{9} \leq 4$ with $y \geq 0$.

5. (10 marks) Let $A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & k & 1 \\ 1 & 0 & 3 \end{bmatrix}$ where k is a real number.

   (a) (3 marks) Find the eigenvalues of A.

   (b) (3 marks) Determine if A is diagonalizable when $k = -2$. If yes, diagonalize it (explanation is not needed); if not, state the reason.

   (c) (4 marks) Find, with careful explanations, all values of k for which A is diagonalizable.

6. (10 marks) Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^3$. Prove or disprove the following statement: $\mathbf{x}+\mathbf{y}$, $\mathbf{y}+\mathbf{z}$ and $\mathbf{z}+\mathbf{x}$ are linearly independent if and only if $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ are linearly independent.

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